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Integrating Machine Learning into Consumer Behaviour Analysis for Insect-Based Protein Supplements

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Abstract—This study explores the application of machine learning techniques to support data-driven marketing for insect-based protein supplements in Thailand, using Brand X as the focal case. Survey data were analysed using three methodological approaches: classification, clustering, and association rule mining. To estimate consumers' likelihood of trying the product, three classification algorithms—Logistic Regression, Decision Tree, and Random Forest—were implemented, with Random Forest achieving the highest performance (76.11% accuracy). K-means clustering identified two distinct consumer profiles: (1) premium- and variety-oriented consumers who emphasise product quality, certifications, and flavour diversity, and (2) price- and safety-conscious consumers who prefer familiar flavours and lower price ranges. FP-Growth and association rule mining further uncovered behavioural patterns, revealing that price concern is the most frequent and influential factor influencing purchasing decisions. The analysis showed that consumers motivated by weight control and those preferring vanilla flavour are often associated with the mid-tier price range (801–1,200 THB). Overall, the findings demonstrate that machine learning offers valuable insights for segmentation, product positioning, and pricing strategies within the emerging insect-based protein market, and can support more targeted and evidence-driven marketing decision-making.

Keywords: Machine Learning, Consumer Segmentation, Insect-Based Protein, Association Rule Mining, K-Means Clustering

I. INTRODUCTION

The global market for protein supplements has experienced notable growth, driven by rising health consciousness, increasing interest in sports nutrition, and the preference for convenient, protein-dense food products [1]. In parallel, insect-derived protein has gained attention as a sustainable and nutrient-rich option, owing to its minimal environmental footprint and highly efficient production processes, as highlighted by the Food and Agriculture Organization of the United Nations (FAO) [2].

Although insect-based protein offers clear nutritional and environmental benefits, its acceptance among consumers

remains comparatively low. Market studies indicate increasing curiosity toward insect-derived protein; however, they also highlight several barriers, including feelings of disgust, concerns regarding safety, limited cultural exposure, and insufficient public understanding [3]. Consumer willingness to incorporate insect-based products into their diets differs widely and is shaped by cultural norms, prior experience, perceived advantages, and psychological readiness to accept such foods [4]. As these products currently occupy a specialised market segment and are often sold at a premium relative to conventional protein sources, identifying potential adopter groups necessitates a data-driven approach to consumer segmentation. To overcome the constraints of conventional segmentation approaches, recent research has increasingly applied machine learning (ML) to capture complex consumer behaviour and reveal latent decision-making patterns [5]. ML methods—particularly clustering, predictive classification, and association rule mining (ARM)—have shown considerable effectiveness in detecting behavioural tendencies and predicting the adoption of innovative nutritional products, including both plant-based and insect-derived protein alternatives [6].

Building on existing findings, this study employs three ML approaches to examine how consumers perceive insect-based protein products. Clustering is applied to identify latent consumer segments among individuals who express interest in such products [7]. Predictive classification models—namely Logistic Regression, Random Forest and Decision Tree—are then used to estimate the likelihood that a consumer would be willing to try insect-derived protein, based on behavioural and attitudinal attributes [8]. Subsequently, association rule mining is utilised to uncover patterns relating to preferences, concerns, and purchase motivations, thereby identifying factor combinations that shape acceptance of insect-based protein [9]. Prior work in market basket analysis demonstrates that association rule techniques are valuable for revealing hidden behavioural associations within consumer datasets [10]. Further literature also supports the use of rule-based and ML-oriented behavioural analysis to inform targeted marketing strategies [11].

Therefore, this paper applies clustering, classification, and association rule mining to a real-world case study of insect-based protein supplementation. By integrating these three ML techniques, the study provides deeper insight into consumer characteristics, key decision drivers, and potential market segments for insect-based protein products—offering practical guidance for product positioning and marketing strategy development within the emerging insect-protein market [12].

II. RELATED WORKS

A. Research Context

The protein supplement industry has grown rapidly as consumers increasingly look for health-oriented, nutrient-dense products that align with more active lifestyles [13]. In Thailand, interest in alternative protein sources has risen, with particular attention given to insect-based options due to their sustainability and nutritional value [14]. Nevertheless, these products encounter several obstacles, including low consumer familiarity, concerns about taste, and varying levels of acceptance [15].

Gaining a detailed understanding of consumer needs, preferences, and purchasing motivations is essential for designing effective marketing strategies and positioning such products in a competitive environment [16]. Insights into what consumers prioritise—such as price, flavour, safety, convenience, and overall product quality—are particularly valuable for informing product development and promotional planning [17].

This study uses ‘Brand X’ as a focal case to examine consumer requirements, motivational drivers, and price sensitivity within the insect-based protein category. The aim is to enhance customer segmentation and provide strategic guidance for decisions relating to product development, pricing structures, and marketing communication.

B. Random Forest Algorithm

Random Forest (RF) is an ensemble-based classification approach that aggregates multiple decision trees to enhance predictive accuracy while reducing the risk of overfitting [18]. It is widely applied in marketing analytics to predict consumer behaviour and to identify influential factors in decision-making processes [19]. Existing studies indicate that this technique performs well with complex behavioural datasets, particularly when attributes such as perceived product quality, safety concerns, and price sensitivity interact [20]. Owing to these strengths, RF is well-suited for examining consumer acceptance patterns in emerging product categories, including insect-based protein supplements [21].

Since this ensemble learning technique combines several decision trees to increase classification accuracy. Although the Random Forest model cannot be represented by a single formula, the algorithm consists of several essential steps and elements:

1) By sampling with replacement, the algorithm uses bagging to generate several subsets of the training data. A distinct bootstrap sample is used to train each decision tree in the forest.

2) Every decision tree is constructed according to the same principles as defined in the decision tree algorithm, such as employing entropy or the Gini index for classification.

3) Selecting a random group of features is a crucial step in splitting nodes in each decision tree.

4) The Random Forest model's final prediction is determined by a majority vote of each tree's predictions.

C. Logistic Regression Algorithm

Logistic Regression is a widely used statistical method for modelling binary outcomes, such as purchase versus non-purchase decisions [22]. The model estimates the likelihood of an event occurring based on a set of independent variables and evaluates the magnitude of their influence [23]. In marketing research, Logistic Regression has been applied to analyse how factors including price, product attributes, and consumer demographics shape purchasing behaviour [24]. Its straightforward structure and high interpretability make it particularly valuable for predicting consumer adoption of new products, including emerging alternatives such as insect-based protein supplements [25].

Equation (1) demonstrates the Logistic Regression model applied in this study.

$$P(y = 1|x) = \frac{1}{1 + \exp(-(\beta_0 + \sum_{j=1}^p \beta_j X_j))} \quad (1)$$

Where the probability that the dependent variable (Y) equals 1 given the input features (X) is denoted by $P(Y = 1 | X)$. β_0 is denoted by the bias term, β_j are the coefficients corresponding to the input features X_j , and exp is the base of the natural logarithm.

D. Decision Tree Algorithm

Decision Tree is a supervised learning method used for both prediction and classification tasks [26]. It constructs an interpretable tree-like model in which data are recursively split based on feature values, highlighting the most influential decision criteria [27]. In marketing analytics, decision trees are widely utilised to segment consumers, analyse behavioural tendencies, and identify key determinants of purchase decisions [28]. Their transparency and visual structure make them particularly suitable for examining the adoption of new products—such as insect-based protein supplements—where understanding consumer priorities, including taste, price, and safety, is essential [29].

Decision Tree models commonly employ entropy as a metric of dataset impurity, expressed as:

$$H(D) = - \sum_{i=1}^c p_i \log_2(p_i) \quad (2)$$

Where p_i represents the proportion of instances within class (i). This measure directs the splitting procedure and assists in determining which attributes exert the greatest influence on consumer decision-making.

E. K-Means Algorithm

K-Means clustering is an unsupervised learning technique that partitions data points into groups according to their similarity [30]. By analysing consumer preferences, behavioural tendencies, and demographic attributes, it supports the identification of distinct customer segments within marketing research [31]. This method is particularly

useful for firms launching new products, as it can uncover clusters of consumers who share comparable needs or purchasing behaviours [32]. In the context of insect-based protein supplements, clustering enables the targeting of specific customer groups, the refinement of product offerings, and the development of tailored marketing strategies [33].

The algorithm minimises the within-cluster variance:

$$J = \sum_{i=1}^k \sum_{x_j \in C_i} \|x_j - \mu_i\|^2 \quad (3)$$

where C_i is the set of points in cluster i and μ_i is the centroid. This ensures that data points are grouped around the nearest cluster mean.

F. Frequent Pattern Growth

Frequent Pattern Growth (FP-Growth) is an efficient method for identifying frequent itemsets within large datasets, as it avoids the need to generate candidate sets [34]. It is extensively applied in marketing analytics to reveal behavioural patterns, including combinations of product preferences, purchasing motivations, and spending tendencies [35]. This technique helps organisations uncover hidden associations among items, thereby supporting more informed decisions regarding product bundling, promotional strategies, and pricing [36]. For emerging products such as insect-based protein supplements, FP-Growth can highlight recurring purchasing patterns and key factors that influence consumer acceptance [37].

Mathematically, the support of an itemset X is defined as:

$$\text{Support}(X) = \frac{\text{Number of transactions containing } X}{\text{Total number of transactions}} \quad (4)$$

G. Association Rule Mining

Association Rule Mining is a data mining approach used to identify relationships between variables in large datasets through the formulation of “if-then” rules [38]. In marketing research, it is applied to analyse co-occurring consumer behaviours, preferences, and purchasing patterns, enabling more targeted promotions and personalised product recommendations [39]. For new products such as insect-based protein supplements, association rules can reveal important links among flavour preferences, price sensitivity, and consumption motivations, thereby supporting firms in crafting effective marketing strategies and optimising product offerings for different consumer segments [40].

Formally, an association rule can be written as:

$$X \Rightarrow Y, X, Y \subseteq I, X \cap Y = \emptyset \quad (6)$$

where I represent the set of all possible items in the dataset. X and Y are itemset, which are subsets of I .

III. METHODOLOGY

A. Research Overview

Figure 1 provides an overview of the conceptual framework adopted in this research. This study initializes by collecting survey data from respondents, focusing on their lifestyle, preferences, and willingness to try alternative protein sources. This step is crucial because it provides the main source of information that the whole study relies on. After gathering the responses, the data is cleaned and pre-

processed by checking for errors, handling missing values, getting rid of irrelevant entries, and organising everything into a clear and consistent format so it can be utilised for analysis.

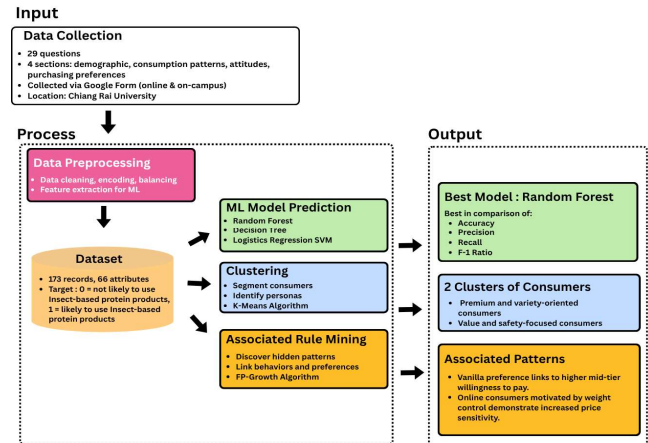


Fig. 1. Conceptual framework

Once the data is properly cleaned, we shift on to the main analytical methods, which consist of three approaches. The first method is classification, which is used to predict the likelihood of respondents trying our product. To achieve this, we use algorithms such as Decision Trees, Random Forest and Logistic Regression to train or classify new data based on past observations. The second approach is clustering, which helps group consumers based on similar traits or behaviours. This allows us to identify different consumer profiles, and we use the K-Means algorithm for this task. The third method is association rule mining, which observes hidden patterns and relationships in the data such as behaviours or preferences that commonly appear together. The FP-growth algorithm is used in this step to efficiently detect frequent itemsets and associations.

B. Data Collection

Data for this study was collected through an online survey divided into four sections: demographics, consumption patterns, attitudes, and purchasing preferences. The demographic section included questions on gender, age, education, occupation, income, region, exercise habits, and prior experience with protein or insect-based protein. A screening question, “Have you drunk protein or insect-based protein before?”, was employed to determine eligibility for subsequent sections. Respondents who answered “Yes” were allowed to proceed to Sections 2 and 3, which focused on their behaviour regarding protein supplement use and their attitudes toward insect-based protein. These sections required respondents to have actual usage experience in order to ensure accurate responses. Conversely, those who selected “No” were directed to Section 4, which avoided inquiries about consumption frequency and preferences. This approach was designed to minimize confusion and reduce survey fatigue, thereby enhancing the quality and validity of the data collected in this research.

Section 2 addressed the importance of protein, label reading, frequency of consumption, types of protein used, and personal goals. Section 3 measured awareness, safety perception, sustainability appeal, concerns, and willingness to give Brand X a try.

Section 4 explored product factors, advertising preferences, promotional rankings, preferred flavours, product forms, acceptable prices, purchasing channels, persuasive messages, and willingness to give feedback, along with open-ended questions for suggestions. This structured attribute design allows the proposed machine learning framework to be applied to other novel or functional food products by adjusting product-specific items while retaining the same analytical structure.

C. Data Preparation

For data cleaning, we started with the 200 survey responses collected. In the question of “Have you drunk protein or insect-based protein before?” question, 27 respondents choose “No”. Since these individuals were not protein consumers and could not provide meaningful information for the consumption and attitude sections, their rows were removed, leaving 173 valid responses for analysis.

The missing values were found in one question from section - 4 and replaced with the mode since the item used a scale of 1 to 5. We then removed unnecessary demographic columns. Although they helped ensure diverse and representative responses, they were not directly relevant to the specific objectives of this study, which focused on product preferences, marketing factors, and consumer attitudes toward insect-based protein. Therefore, we kept only the variables related to product and marketing preferences. Column names were simplified for readability, and descriptive labels from rating questions were removed so that only numerical values remained. These steps ensured the dataset was consistent, clean and ready for further analysis.

Following the cleaning process, the data was transformed so it could be used for modelling. Categorical columns were converted into numerical form using binary encoding (one-hot), ordinal, or nominal encoding, depending on the type of variable. We also selected “Likelihood to try Brand X” as the target variable and changed it into a binary value, where high likelihood was coded as 1 (likely) and low likelihood as 0 (not likely). These steps strengthened the dataset in a fully numerical and structured format for machine learning analysis.

D. Clustering with K-Means

For clustering, we focused only on respondents who were likely to try Brand X. After encoding the target variable (“Likelihood to Try Brand X”) as 0 and 1, all entries with 0 were removed, eliminating 60 rows and leaving 113 valid cases. This filtered dataset was then analysed using the K-Means algorithm, allowing us to group fascinated consumers based on similar preferences and behaviours. By clustering only potential users, the results more accurately revealed unique consumer segments and the patterns associated with them.

We clustered the consumers into two groups for two main reasons. From a data perspective, the remaining dataset was relatively small (113 cases), and choosing two clusters helped prevent over-segmentation, which can lead to unstable or irrelevant group structures. From a marketing perspective, segmenting potential customers into two broad groups is practical and interpretable, allowing marketers to easily differentiate between unique consumer profiles such as value-driven buyers versus convenience-driven buyers and

develop targeted strategies accordingly without confusion. The dataset was intentionally segmented into two clusters because preliminary exploration showed that additional clusters produced unstable and weakly differentiated groups, while a two-cluster solution provided clearer strategic value by separating premium-oriented consumers from price- and safety-focused consumers in an emerging market context.

E. Association Rule Mining

For association rule mining, we selected only binary variables, including preferred flavours, price ranges, purchasing channels, protein types, concerns, marketing messages, and other product-related attributes. These features were encoded as 0 or 1, indicating whether each condition applied to a respondent. Using this transformed dataset, we utilized the FP-growth algorithm in RapidMiner to determine meaningful relationships between consumer behaviours and preferences. The minimum confidence threshold was set to 0.7 to retain only strong association rules, meaning that the consequent occurred in at least 70% of cases where the antecedent was present, while avoiding an excessive number of weak or uninformative rules.

F. Machine Learning Models

Machine Learning procedures were conducted using RapidMiner’s Altair AI Studio, focusing on four supervised algorithms: Random Forest, Decision Tree and Logistic Regression. The cleaned dataset was imported, variable roles were assigned, and the data was split into training and testing sets using a 70-30 ratio. Each model was trained using its default parameters first, and then parameters were adjusted step-by-step to improve performance. Accuracy, precision, recall, and F1 score were used to compare models, and the best-performing parameter settings were selected based on achieving the highest F1 score.

IV. RESULTS

In this section, the results of each machine learning process will be discussed in the following order: classification results, clustering results, and frequent itemset and association rule results.

A. Classification Results

As mentioned in the methodology, three different classification algorithms—Logistic Regression, Decision Tree, and Random Forest were employed to predict the likelihood of an individual becoming a customer. Table I shows the parameters used for each model.

TABLE I. PARAMETER SETTINGS OF CLASSIFICATION MODELS

Random Forest
Number of trees = 100, maximal depth = 100, voting strategy = confidence vote, criterion = accuracy, apply guess subset ratio = Yes, apply enable parallel execution = Yes, apply random split = Yes
Logistic Regression
Kernel type = multiquadric, kernel sigma1 = infinity, kernel shift = 1.0., kernel cache = 200, C = 1.0, convergence epsilon = 0.001, max iterations = 100000, apply scale = Yes
Decision Tree
Criterion = accuracy, Maximal depth = 100, apply prepruning = Yes, minimal gain = 0.01, minimal leaf size = 2, minimal size for split = 4, number of prepruning alternatives = 3

As shown in Table II, Random Forest achieved the highest accuracy (76.11%), precision (75.86%), recall (76.33%), and f1-score (74.70%), outperforming Logistic Regression and Decision Tree algorithms. Logistic Regression achieved the second highest performance with 72.29% accuracy, while Decision Tree ranked third in performance with 71.54% accuracy. The results indicate that the Random Forest model is the most effective at predicting the target outcome—Likelihood to try Brand X.

TABLE II. PERFORMANCE COMPARISON OF CLASSIFICATION MODELS

Model	Model performance comparison			
	Accuracy	Precision	Recall	F1-score
Random Forest	76.11%	75.86%	76.33%	74.70%
Logistic Regression	72.29%	72.25%	72.25%	72.25%
Decision Tree	71.54%	70.8%	71.7%	71.1%

B. Clustering Results

K-means algorithm was employed for customer segmentation, which resulted in 2 clusters of customers with distinct behaviours and preferences. The clustering process focused exclusively on respondents who expressed an interest in trying Brand X, i.e., the rows with value “1” in the “Likelihood to try Brand X” column. Fig. 1 represents the two clusters resulting from the K-means algorithm.

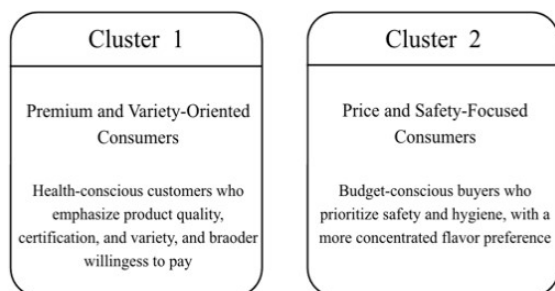


Fig.2. Customer Clusters

Cluster 1 represents the “Premium and Variety-Oriented Consumers”. Respondents in this cluster reported higher average scores on the importance of protein intake, product certifications, ingredient transparency, serving size, flavour, and packaging design. They have higher receptiveness towards marketing activities, particularly social media advertising, tasting events, and promotional campaigns. Flavour preferences were distributed more evenly across all the flavours available: Vanilla, Matcha, Original, and Chocolate. In terms of price, this group showed a broader willingness to pay, including mid to higher price ranges (500–1,200 THB).

Cluster 2 represents the “Price and Safety-Focused Consumers”. Compared to Cluster 1, this cluster expressed stronger concerns for safety and hygiene. The flavour preference was more concentrated in this cluster with Vanilla emerging as the dominant choice. The price range “below 500 THB” was the most favoured, indicating the price sensitivity of this cluster. Additionally, this group was less influenced by marketing and promotional activities.

C. Frequent Itemset and Association Rule Results

The Frequent Pattern Growth (FP-Growth) algorithm was used to identify frequent item sets that reveal consumer behaviour patterns related to Brand X insect-based protein products. Table II represents the frequent consumer behaviour and preferences identified by FP-Growth algorithm.

TABLE II. FREQUENT ITEMSETS IDENTIFIED BY FP-GROWTH

No.	Size	Frequent Itemset	Support
1	1	Price Concern	0.855
2	1	Drink Protein for Weight Control	0.734
3	1	801 –1,200 THB	0.723
4	2	Price Concern + Drink Protein for Weight Control	0.624
5	2	Drink Protein for Weight Control + 801 –1,200 THB	0.549
6	2	801 –1,200 THB + Preferred flavour Vanilla	0.457

The most frequent item was Price Concern with 85.5% support. This means that affordability is the most concerned issue across respondents. The next most frequent itemsets were Drink Protein for Weight Control (support = 0.734) and Willingness to pay 801–1,200 THB (support = 0.723). These results suggest that weight management is the dominant consumption motivation, and a significant portion of consumers preferred the 801–1,200 THB price range.

Combined itemsets provide further insights into consumer segments. The pairing of Price Concern and Drink Protein for Weight Control (support = 0.624) suggests that consumers who consume protein for weight control are simultaneously highly price-sensitive. Weight-control-oriented consumers also prefer the 801–1,200 THB price range, suggested by the itemset “Drink Protein for Weight Control + 801 –1,200 THB” (support 0.549). Flavour preferences also interact with pricing; the itemset “801–1,200 THB + Preferred Flavour Vanilla” (support = 0.457) highlights that consumers with vanilla flavour preference prefer the 801–1,200 THB price range.

TABLE III. THE TWO-ITEM ASSOCIATION RULES

No.	Rules (A B)	Support	Conf	Lift
1	Drink Protein for Weight Control -> Price Concern	0.624	0.850	0.994
2	801–1,200 THB -> Price Concern	0.595	0.824	0.963
3	Drink Protein For Weight Control -> 801–1,200 THB	0.549	0.748	1.035
4	Marketing message: High protein & nutrition -> Price Concern	0.474	0.837	0.978
5	Preferred flavor Vanilla -> 801–1,200 THB	0.457	0.823	1.139

Table III represents the association rules between two items. The analysis of two-item association rules indicates that price concern consistently co-occurs with consumption motivations and product preferences. Specifically, consumers who consume protein for weight control demonstrate a strong likelihood of being price-sensitive (support = 62.4%, confidence = 85.0%). Similarly, individuals willing to pay

within the mid-price range (801–1,200 THB) are also associated with price concern (support = 59.5%, confidence = 82.4%). This suggests that even consumers prepared to spend at a moderate level remain attentive to affordability considerations. Marketing messages highlighting “high protein and nutrition” were also linked to concerns about price (support = 47.4%, confidence = 83.7%), suggesting that nutritional claims alone may not reduce consumer price sensitivity. Another notable pattern showed that a preference for vanilla flavour tended to align with a willingness to pay within the 801–1,200 THB range (support = 45.7%, confidence = 82.3%, lift = 1.14).

V. CONCLUSION

This study applied classification, clustering, frequent itemset and association rule mining to analyse consumer attitudes and behaviours toward Brand X, an insect-based protein product. Among the classification models, Random Forest achieved the best overall performance, making it the most reliable tool for predicting which consumers are likely to try Brand X. K-means clustering identified two distinct consumer groups: “Premium and Variety-Oriented Consumers”, and “Price and Safety-Focused Consumers”. The clusters suggest that Brand X may benefit from offering both premium-positioned and budget-friendly options. Frequent itemset mining and association rule analysis further enhanced the understanding of purchasing motivations and preference patterns. Price concern emerged as the most common factor among respondents, and weight control was identified as a major consumption motivation. Several rules demonstrated a strong connection between price sensitivity and both health-related motivations and flavour preferences. Notably, the association between vanilla flavour preference and the 801–1,200 THB price range suggests the presence of a segment with mid-tier spending tendencies.

The limitation of this study is the small dataset of 200 survey responses, which may limit the representativeness and robustness of the results. The data also rely on self-reported responses, which can introduce bias if participants do not accurately reflect their true attitudes or behaviours. In addition, the survey captures consumer perceptions at a single point in time. Preferences may change over time due to market trends, exposure to insect-based protein, or economic conditions, which limits the long-term generalizability of the results.

ACKNOWLEDGEMENT

This research was supported by the Center of Excellence in AI and Emerging Technology and School of Applied Digital Technology, Mae Fah Luang University, which provided funding, high-performance facilities, and laboratory equipment.

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